AI FUNDAMENTALS · END-TERM PROJECT

TULPAR-MAN

TEN NIGHTS OVER ASTANA

A ten-issue comic that teaches the whole AI Fundamentals course

Team 67 · Alua · Eldar · Zeynep · Aruzhan



WHY A COMIC

A VILLAIN YOU REMEMBER IS A FORMULA YOU REMEMBER.

Companion to our e-book

The book explains in depth. The comic makes each idea stick in one picture.

One week, one issue

Ten issues follow the ten weeks of the syllabus, in order.

Real numbers

Every formula in a panel comes from our lecture examples.

HOW THE COMIC IS BUILT

ORIGIN → ISSUES #01–#10 · WEEKS 1–10 → DAWN

INSIDE EVERY ISSUE

1 · The attack

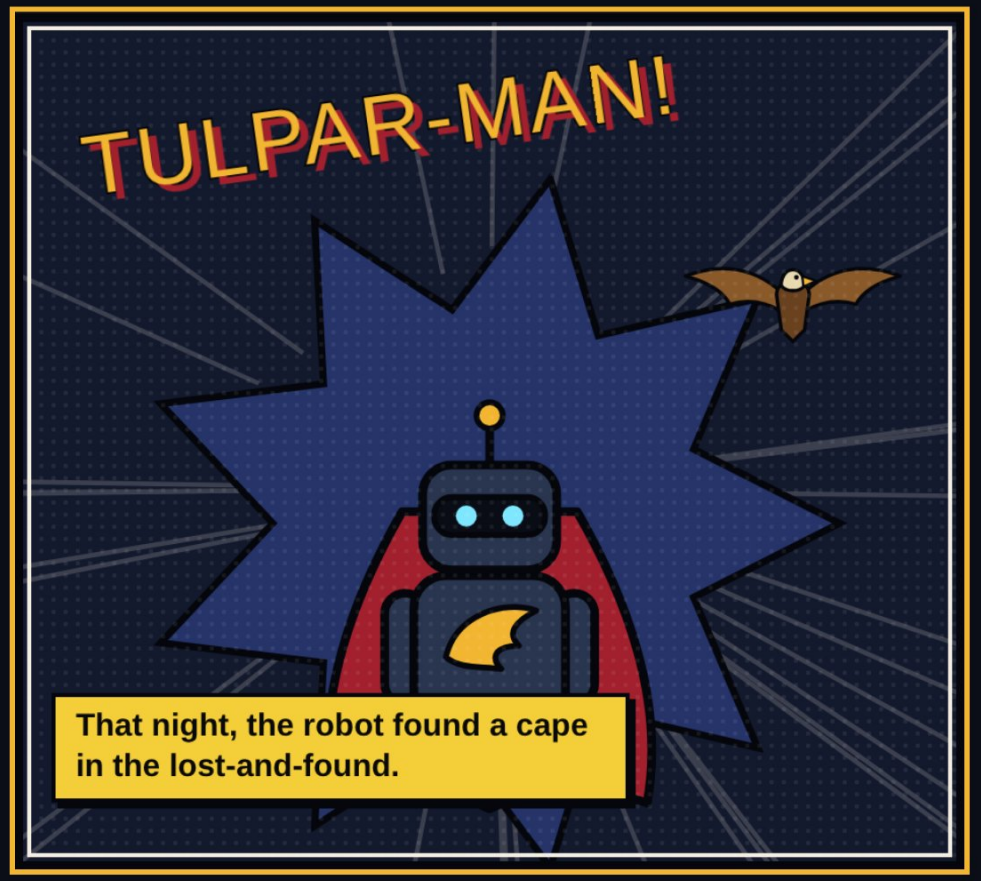
A wide panel: a villain breaks Astana with a bad algorithm.

2 · The idea

Rows of two panels: Tulpar-Man thinks, draws the diagram and wins.

3 · The Tulpar File

A one-paragraph exam summary under the panels.



OUR CAST

Tulpar-Man

A university delivery robot who learns to think. Named after the tulpar, the winged horse of Kazakh legends.

Bürkit the golden eagle

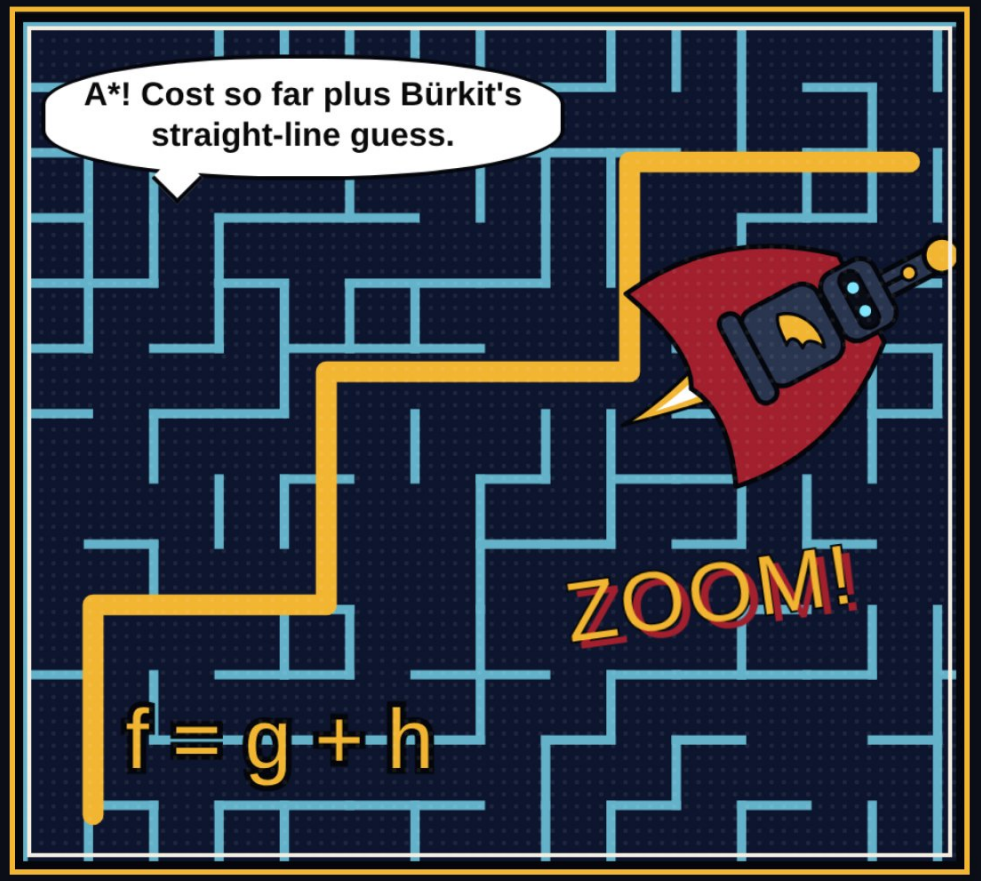
The mentor. He sees the city from above, like a straight-line heuristic.

Astana as the stage

Bäiterek, Khan Shatyr, EXPO and a game of togyz kumalak.

TEN WEEKS, TEN VILLAINS

Week	Course topic	Villain	Defeated by
1	Agents and environments	Reflex Rex	PEAS, rational agents
2	Uninformed search, A*, heuristics	The Labyrinth	A*, admissible heuristic
3	Local search and CSP	Local Maximus	Annealing, restarts, CSP
4	Adversarial search, game trees	Count Zero-Sum	Minimax, alpha–beta
5	MDPs and reinforcement learning	Black Ice	Bellman, Q-learning
6	Probability, Bayesian networks	Baron Base-Rate	Bayes' rule
7	HMMs, sequential inference	The Hidden Hand	Filtering, Viterbi
8	Rational decisions, classical ML	The Overfitter	Recall, expected cost
9	Perceptrons, neural networks	XOR the Unseparable	A hidden layer
10	LLMs, generative AI, AI safety	Doctor Hallucination	Fact-checks, AI law



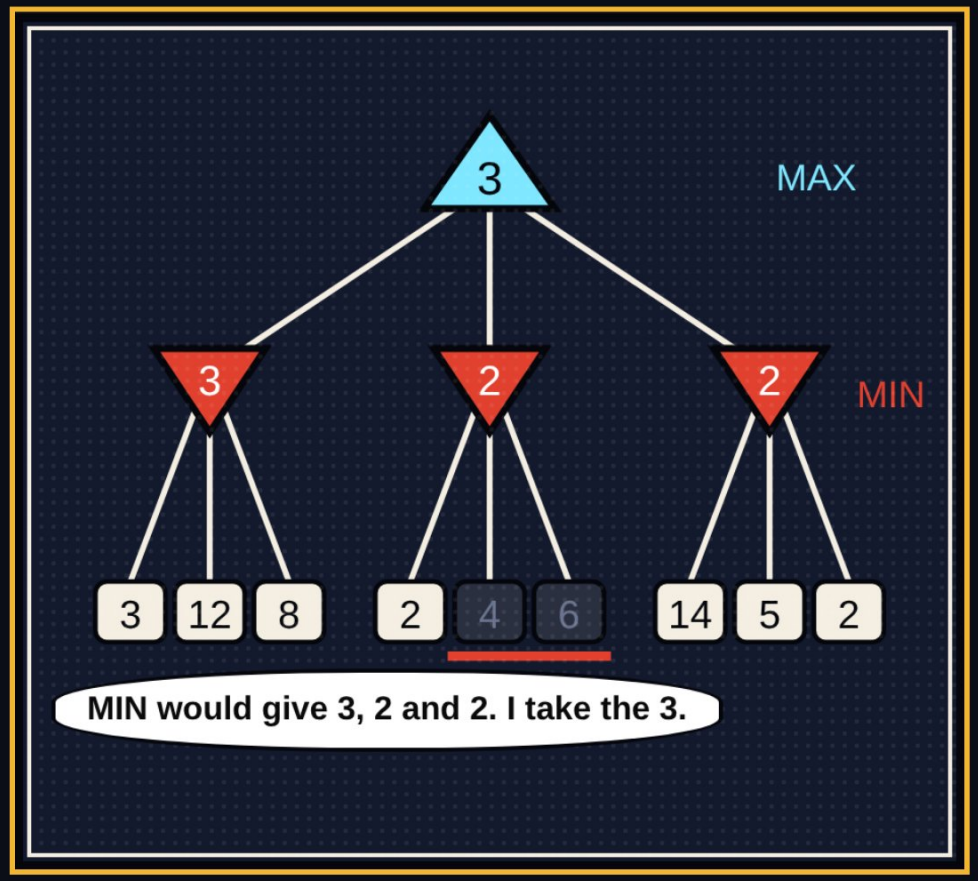
ISSUE 02 · WEEK 2

THE LABYRINTH

The villain hides Bürkit in a maze and boasts that BFS will take all night.

Tulpar-Man uses A*: the cost so far plus the eagle's straight-line guess, which never overestimates.

F(N) = G(N) + H(N)



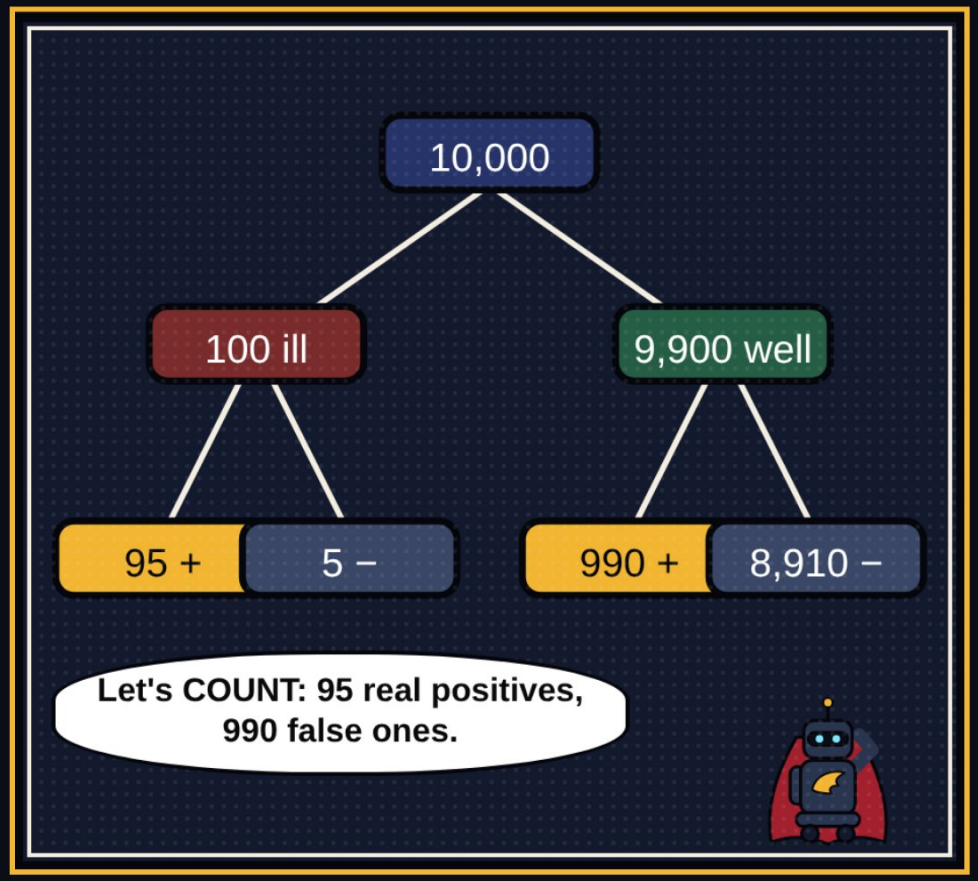
ISSUE 04 · WEEK 4

COUNT ZERO-SUM

On the roof of Khan Shatyr, the villain challenges our hero to togyz kumalak.

MIN takes 3, 2 and 2; MAX takes the 3. Alpha–beta never looks at 4 and 6, and the answer stays the same.

PRUNE WHEN α ≥ β



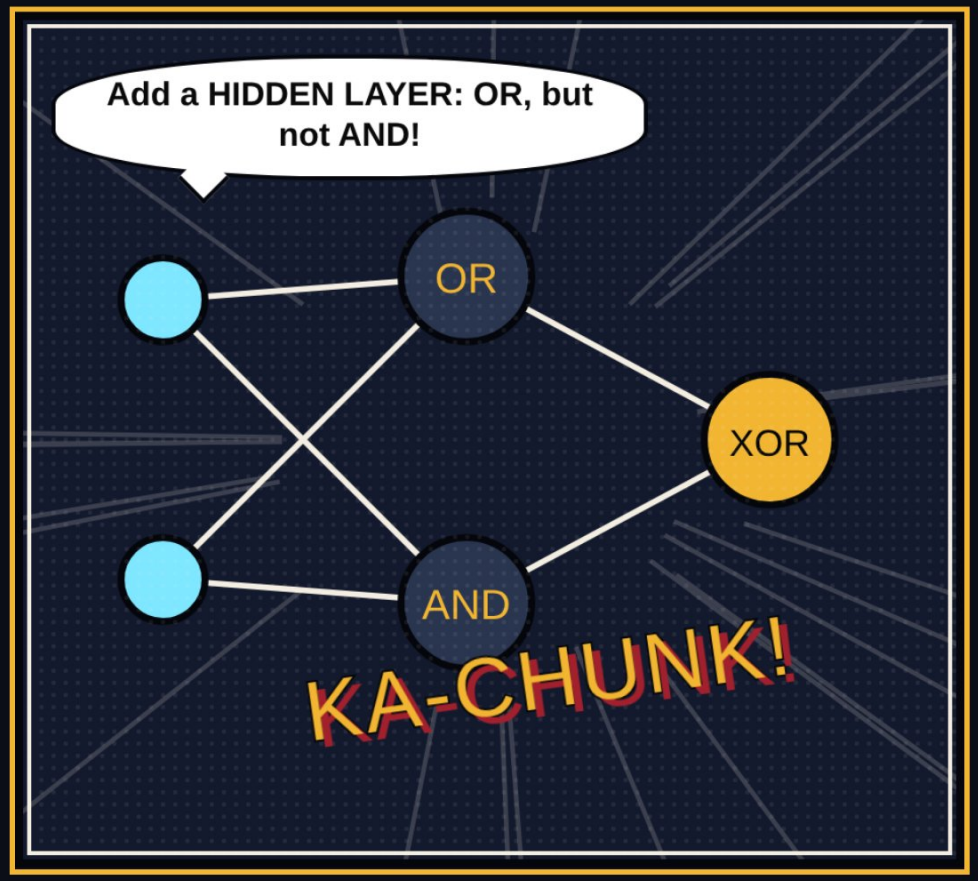
ISSUE 06 · WEEK 6

BARON BASE-RATE

In the fog, the villain spreads panic: the test is 95% accurate, so everyone who tests positive must be ill.

Tulpar-Man counts instead: 95 true positives against 990 false ones.

P(ILL | +) = 95 / 1085
≈ 8.8%



ISSUE 09 · WEEK 9

XOR THE UNSEPARABLE

The villain dares the hero to split his points with one straight line. One perceptron cannot do it.

A hidden layer can: one neuron computes OR, one computes AND, and the output says “OR but not AND”.

$$z = W \cdot x + b$$

EVERY SYLLABUS SUBTOPIC IS DRAWN

10 issues · 58 panels · 40 subtopics, each in its own panel

PEAS · Environment types · Four agent types · BFS · DFS · UCS · A* · Admissible heuristics · Hill climbing · Random restarts · Simulated annealing · CSP · Minimax · Alpha–beta · Expectimax · Depth limit, horizon effect · MDP and policy · Bellman equation · Value iteration · ε-greedy, discount · Q-learning · Conditional probability · Bayes' rule · Bayesian networks · Markov chains · HMM filtering · Prediction, smoothing · Viterbi · Expected utility · Linear, logistic regression · Naive Bayes · K-means · Confusion matrix, recall · Perceptron rule · XOR, hidden layers · Backpropagation · LLMs, temperature · GANs, diffusion · AI safety · Kazakhstan's AI Law

How we made it, and how we checked it

Our process

1. Read the syllabus and lecture slides for weeks 1–8.
2. Wrote the script: one villain per week.
3. Drew every panel as code (SVG), so diagrams stay exact.
4. Checked each number against the slides and recalculated it in Python.
5. Tested the comic on phones and laptops.

AI use, declared

We used Claude (Anthropic) to draft the script, draw the artwork and lay out the pages.

We checked the content, corrected it, and can explain every panel ourselves. The declaration is in the comic's credits and in the e-book.



Scan to read the comic

READ IT NOW

The comic

karassvo.github.io/Tulpar/comic.html

The e-book

karassvo.github.io/Tulpar

No login or install needed; both work on a phone. The README in our GitHub repository explains how to open them offline.

THANK YOU! QUESTIONS?

Alua
Origin, issues #01–#02

Eldar
Issues #03–#05

Zeynep
Issues #06–#07

Aruzhan
Issues #08–#10, Dawn

Team 67 · Tulpar-Man: Ten Nights Over Astana · AI Fundamentals 2026–2027